SEMESTER-VI

URA601: ROBOTIC-SYSTEMS SIMULATION				
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	1	0	4	3.0

Course Objective: The course on robotic systems simulation aims to teach students to program and control robotic systems using ROS. By the end of the course, students should be able to create and visualize URDF and xacro models, simulate robots using Gazebo and Coppeliasim, plan and navigate robots using Moveit and navigation stack, and control robots using joint controllers and visualization in RViz. Finally, students should be able to integrate sensors and actuators with robotic systems using an embedded system.

Syllabus

Introduction to programming (C++): Basic program structure, variables, types, constants, operators, statements and flow control, strings, functions: pass by value and pass by reference, function declaration, arrays, file I/O, vectors, structures and classes, serialization and deserialization of classes, subclasses, and inheritance, polymorphism, coding guidelines.

Introduction to ROS: ROS1 and ROS2, ROS filesystem, nodes, messages, topics, service, bags, ROS master, 3D robot modeling in ROS: create, explain, and visualize URDF and xacro robot model, manipulator and differential drive wheel mobile robot.

Basic Robot simulation: simulation of the manipulator and differential drive in Gazebo and Coppeliasim, Planning and Navigation: SLAM map building, motion planning of robots using ROS-Moveit, ROS navigation stack, ROS controller and visualization: joint controller, visualization using RViz, teleoperation,

Advanced Robot simulation: motion planning, collision, manipulation, and grasping with ROS-Moveit, ROS with MATLAB-SIMULINK: configuration of ROS and MATLAB, controller in MATLAB-SIMULINK, ROS for Industrial Robots: Industrial robot URDF and its control.

I/O Boards, Sensors, and Actuators: Arduino-ROS and RaspeberryPI-ROS for LEDs, accelerometers, ultrasonic sensors, and actuators. Introduction and examples using NVIDIA-ROS.

Laboratory Work

Lab 1: Introduction to C++ programming: write basic programs using variables, types, operators, and functions.

Lab 2: Arrays, vectors, and structures: write programs that use arrays, vectors, and structures for data storage and manipulation.

Lab 3: Object-Oriented Programming: create classes and implement inheritance, polymorphism, serialization, and deserialization of classes.

Lab 4: 3D Robot Modeling with ROS: create and visualize URDF and xacro models of robots using RViz and Gazebo.

Lab 5: Robot simulation in Gazebo and Coppeliasim: simulate robots with different configurations and control methods.

Lab 6: Motion Planning and Navigation with ROS: plan and navigate robots using Moveit and navigation stack.

Lab 7: Robot Control with ROS: control robots using joint controllers and visualization in RViz, and perform teleoperation. Lab 8: Advanced Robot Simulation: implement motion planning, collision detection, manipulation, and grasping with ROS Moveit. Lab 9: Integration of Sensors and Actuators: interface sensors and actuators with Arduino-ROS and RaspberryPI-ROS and control LEDs, accelerometers, and ultrasonic sensors. Lab 10: Final Project: apply the skills learned in the course to design and implement a robotic system with a specific task or application.
Course Learning Objectives (CLO) The students will be able to: 1. program in C++ and implement functions, arrays, vectors, and classes for object-oriented programming. 2. create and simulate robots using URDF, xacro, Gazebo, and Coppeliasim. 3. plan and navigate robots using ROS Moveit and navigation stack. 4. control robots using joint controllers, visualization in RViz, and teleoperation. 5. integrate sensors and actuators with robotic systems using embedded systems.
Text Books 1. Mastering ROS for Robot Programming, Lentil Joseph and Jonathan Cacace, Packt Publications, 2nd ed. 2018.
Reference Books & Websites 1. Robot Operating System: The complete reference, Anis Koubaa, Springer, vol.6, 2021 2. http://wiki.ros.org/ROS/Tutorials

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	Sessional tests/assignments on software	30
2	Projects on development of robot model, simulation, motion planning, guidance control, etc., as relevant to the project. With technical reports of each.	70

NB: 50% pass marks. Tests and projects on software will be open-book examinations.